MIL-D-48046A (MU)
16 August 1973
SUPERSEDING
MIL-D-48046 (MU)
4 February 1972

MILITARY SPECIFICATION

DETONATOR, STAB: XM91

LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of detonator designated as Detonator, Stab: XM91.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550	- Ammunition, General Specification for

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	- Quality Assurance Terms and Definitions

FSC: 1375

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

MIL-STD-1168 - Lot Numbering of Ammunition

MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. Army Munitions Command

9229739	- Detonator, Stab, XM91
9258182	- Box, Set-up, Packing Ammunition, Inner, for Small Explosive Components
9258183	- Box, Set-up, Packing Ammunition, Outer, for Small Explosive Components
9258184	- Box, Packing, Ammunition for Small Explosive Components
8802085	- Fixture, Testing
8842964	- Adapter, Ball
9202648	- Fixture, Test
8827829	- Bushing
8841278	- Firing Pin
9202649	- Holder, Firing Pin
9202650	- Disc, Steel

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

. Title 49 - Transporatation 0-190

(The Code of Federal Regulations is available from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. Orders should specify, "49 CFR 0-190 (latest revision)".)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Detonators.-The detonators shall comply with all requirements specified on Dwg. 9229739, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Functioning.

3.3.1 Functioning test.-The detonators shall function satisfactorily as evidenced by complete destruction when tested as specified in 4.4.2.1.

3.3.2 Waterproofness.-The detonators shall be waterproof as evidenced by functioning after water immersion when tested as specified in 4.4.2.2.

3.4 Detonator sticking to packaging.-The detonators shall not stick to the onion skin or spacer in the inner set-up box when tested as specified in 4.4.3.

3.5 Workmanship.-All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All marking and stamping shall be neat and sharply defined.

3.6 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (See 6.5).

3.7 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements when such material is purchased by the contractor or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificates shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

4.2.1 Submission.-The contractor shall submit a first article quantity of four hundred (400) detonator assemblies as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1).

All parts and completed assemblies submitted shall have been produced by the Contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process (including heat treatment procedures and equipment), material used, drawing specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspection to be performed.-Assemblies will be subjected by the Government to the following tests.

4.2.2.1 Functioning test.-Two hundred (200) detonator assemblies shall be subjected to this test. If two (2) or more assemblies fail to comply with the applicable requirement, the first article quantity shall be rejected and the contractor shall take the necessary corrective action and submit a new first article quantity.

4.2.2.2 Waterproofness.-Two hundred (200) detonators assemblies shall be subjected to this test. If six (6) or more assemblies fail to comply with the applicable requirement, the first article quantity shall be rejected and the contractor shall take the necessary corrective action and submit a new first article quantity.

4.2.3 Rejection.-In the event of rejection, the Government reserves the right to require the contractor to take corrective

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

action and submit a new first article quantity. Until the first article quantity is accepted, the contractor is not authorized by the Government to proceed with regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions.

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process in accordance with the same drawing, same drawing revision, same specification and same specification revision, and complies with the provisions for submission of product as specified in MIL-STD-105. Change to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

a. Metal parts of one lot interfix number from one manufacturer.

b. Primer mixture consisting of batches blended by the same process. Each batch shall be composed of the same ingredients, each of which is to be from one lot interfix from one manufacturer.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plan for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Individual defects shall have an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

4.3.2.1 Detonator, Stab: XM91, prior to loading (See Dwg. 9229739).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:			
	1. Cup cut, split or cracked	Visual	01001
Major: None defined			
Minor			
	201. Evidence of poor workmanship (See 3.5)	Visual	01002

4.3.2.2 Detonator, Stab: XM91 (See Dwg. 9229739).

Categories	Defects	Method of Inspection	Code No.
Critical:			
	1. Caked explosive on exterior	Visual	02001
	2. Explosive exposed	Visual	02002
Major:			
	101. Total length, maximum (max)	Gage	02003
	102. Outside diameter	Gage	02004
	103. Crimp radius (max)	Gage	02005
	104. Paint on wrong end or paint on both ends or no paint	Visual	02011
Minor			
	201. Distortion	Visual	02006
	202. Disc wrinkled or folded	Visual	02007
	203. Crimp not full 360 degrees	Visual	02008
	204. Crack at crimp	Visual	02009
	205. Evidence of poor workmanship (See 3.5)	Visual	02010

4.3.2.3 Box, set-up, inner, after loading and prior to closing slide cover (See Dwg. 9258182).-

Categories	Defects	Method of Inspection	Code No.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

Critical:			
	1. Felt cushion missing	Visual	12001
	2. Filler missing	Visual	12002
	3. Stiffener missing	Visual	12003
	4. Slide cover assembled reversed	Visual	12004
Major			
	101. Release paper missing or improperly assembled	Visual	12005
Minor			
	201. Marking missing, incomplete or illegible	Visual	12006
	202. Evidence of poor workmanship (See 3.5)	Visual	12007

4.3.2.4 Box, set-up, outer, with inner set-up boxes assembled and prior to assembling top and sealing strip (See Dwg. 9258183).-

Categories	Defects	Method of Inspection	Code No.
Critical:			
	1. Red surfaces on inner set-up boxes do not alternate on same side of outer set-up box	Visual	13001
	2. Black surfaces on inner set-up boxes do not alternate on same side of outer set-up box	Visual	13002
Major:	None defined		
Minor			
	201. Marking misleading, incomplete or illegible	Visual	13003
	202. Evidence of poor workmanship (See 3.5)	Visual	13004

4.3.2.5 Box, packing, ammunition, for small explosive components, prior to assembling cover and strapping (See Dwg. 9258184).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

Major			
101.	Bags containing set-up boxes damaged	Visual	14001
102.	Cushioning material incomplete or missing from around outer set-up boxes	Visual	14002
Minor			
201.	Evidence of poor workmanship (See 3.5)	Visual	14003

4.3.2.6 Box, packing ammunition for small explosive components (See Dwg. 9258184).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major			
101.	Strapping missing, broken, loose, or improperly fastened	Visual/Manual	15001
102.	Box damaged	Visual	15002
103.	Marking missing, incomplete or illegible	Visual	15003
Minor:	None defined		

4.3.3 Testing .

4.3.3.1 Moisture content of primer mixture (See Dwg. 9229739) Major Defect, Code No. 09001.-At least one six (6) - gram sample of primer mixture shall be selected for the moisture content determination from that quantity to be used in each day's production of detonators. Sample of primer mixture shall be selected at the site of and immediately before loading. The incidence of moisture in the sample in excess of the applicable drawing requirements shall be cause for rejection of that quantity of detonators represented by the sample. Testing shall be as specified in 4.4.1.

4.3.3.2 Functioning test, Major Defect, Code No. 10001.-A sample size of 125 detonators shall be selected for this test. If two (2) or more detonators fail to function the lot shall be rejected. Testing shall be as specified in 4.4.2.1 using equipment in accordance with 4.3.4. (See 2.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

4.3.3.3 Waterproofness, Major defect, Code No. 11001.-A sample of 80 detonators shall be selected for this test. If two (2) or more detonators fail this test the lot shall be rejected.

4.3.3.4 Detonator sticking to packaging (See 3.4) Major Defect Code No. 16001.-A sample of twenty-five (25) inner set-up boxes from each lot shall be selected for this inspection. No more than one (1) inner set-up box shall be selected from each outer set-up box opened, and in addition, the samples shall be selected a minimum of forty-eight (48) hours after assembly of the inner set-up boxes into the outer set-up boxes. If twenty-one (21) or more detonators from the sample stick to either the release paper or inner set-up box, the lot shall be rejected. This inspection shall be performed as specified in 4.4.3, using equipment satisfactory to Contracting Officer.

4.3.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.3.2 and in 4.3.3 (See 6.3).

4.4 Test methods and procedures .

4.4.1 Test of primer mixture.

4.4.1.1 Moisture content (See 4.3.3.1).-Transfer an accurately weighted portion of approximately one gram of the primer mixture to a tared aluminum weighing dish provided with a teflon cover while both the dish and cover are on the balance pan. Place the dish and contents with the cover removed, in an oven maintained at 60° ± 5°C for 2 hours. The cover shall be placed in a desiccator during the drying period. After drying, transfer the dish and contents quickly to a desiccator and cool to room temperature. Replace the plastic cover and weigh. Calculate the loss in weight of the dish cover and contents to percent moisture in the sample.

4.4.1.2 Composition (See 7548654).-The analysis of the composition shall be performed as follows:

4.4.1.2.1 Barium nitrate.-Transfer an accurately weighed portion of approximately 0.5 gram of the dried sample (dried as described in 4.4.1.1) to a tared medium porosity fritted glass crucible of 30 milliliter (ml) capacity. Record the weight of crucible and sample as Weight A. Add 3 ml of distilled water saturated with lead azide at a temperature of 5 ± 2°C. Swirl

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

for 30 seconds, then apply suction. Repeat this procedure two times, making a total of three 3-ml extractions. Using an eye dropper, transfer to the crucible enough alcohol to thoroughly wet the sample and wash the crucible walls. Aspirate until the odor of alcohol is no longer detectable and dry the crucible in an over for 30 minutes at 60° + 5°C. Cool in a desiccator and weigh. Record this weight at Weight B. Reserve the crucible and residue for the normal or basic lead styphnate determination. Correct for the percentage of normal or basic lead styphnate in the water extract as follows: Transfer the filtrate and washings quantitatively to a 100-ml calibrated volumetric flask, washing the beaker thoroughly with distilled water and dilute to the mark. Using a calibrated pipette, transfer a 10-ml aliquot of this solution to a second 100-ml calibrated volumetric flask and dilute to the mark with distilled water. Transfer a portion of this solution to a corex glass spectrophotometric cell (See 6.3) having a width of approximately 1 centimeter (cm) and determine its optical density at a wave length of 410 millimicrons. Use as reference a similar cell containing the same proportion of water saturated with lead azide and alcohol as the sample solution. Since the cell which holds the lead styphnate solution is not identical with the reference cell, correct the optical density obtained as described above for the difference in the amount of light which the two cells scatter and absorb. To do this, fill both cells with a solution of water and alcohol, and measure at a wave length of 410 millimicrons, the optical density of the cell which originally contained the normal lead styphnate solution. Calculate the percentage of normal lead styphnate in the filtrate on a dry basis as follows:

Normal lead styphnate found in barium nitrate filtrate,

$$\text{Percent Pn} = \frac{2.94\ (A-B)}{WD}$$

Basic lead styphnate found in barium nitrate filtrate,

$$\text{Percent Pb} = \frac{4.48\ (A-B)}{WD}$$

where:

A = Optical density of normal lead styphnate solution.
B = Optical density of water and alcohol solution.
D = Width of corex cell, cm
W = Weight of dried sample.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

Calculate the percentage of barium nitrate in the sample as follows:

$$\text{Barium nitrate, percent} = \frac{100\ (A-B)}{W} - (\text{Pb or Pn})$$

where:

W = Weight of the dried sample.
A = Weight of crucible and sample.
B = Weight of the crucible and sample after extractions with water saturated with lead azide.
Pn = Percentage of normal lead styphnate found in barium nitrate filtrate.
Pb = Percentage of basic lead styphnate found in barium nitrate filtrate.

4.4.1.2.2 Basic or normal lead styphnate.-The residue obtained in the determination of barium nitrate shall be extracted with six-ml portions of ammonium acetate solution at about 25°C, (hot ammonium acetate shall not be used as it dissolves antimony sulfide to some extent) agitating each portion for about 30 seconds and allowing each portion to remain in contact with the residue for approximately 2 minutes. The residue shall be washed with water, saturated with antimony sulfide, until the washings coming through the crucible appears colorless. The filtrate and washings shall be quantitatively transferred to a 250-ml calibrated volumetric flask, the beaker washed thoroughly with distilled water and diluted to the mark. Using a calibrated pipette, a 2-ml aliquot of this solution shall be transferred to a 50-ml calibrated volumetric flask in the case of basic lead styphnate, or a 2-ml aliquot of the solution shall be transferred to a 100-ml calibrated volumetric flask in the case of normal lead styphnate, and diluted to the mark with distilled water. A portion of this solution shall be transferred to a corex glass spectrophotometric cell having a width of approximately one centimeter (cm) and its optical density determined at a wave length of 410 millimicrons. As a reference, a similar cell containing ammonium acetate solution of the same concentration as the solution of the specimen shall be used. Since the cell which holds the basic lead styphnate is not identical with the reference cell, the optical density obtained shall be corrected as described above for the difference in the amount of light which the two cells scatter and absorb. This shall be done by filling both cells with ammonium acetate solution and measuinrg at a wave length of 410 millicicrons, the optical density of the cell which originally contained the lead styphnate solution. The percentage of normal or basic lead styphnate in the specimen shall be calculated as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

$$\text{Percentage basic lead styphnate} = \frac{28\ (A-B)}{WD} \pm Pb$$

$$\text{Percentage normal lead styphnate} = \frac{36.75\ (A-B)}{WD} \pm Pn$$

where:

A = Optical density of normal or basic lead styphnate.
B = Optical density of ammonium acetate solution.
D = Width of corex cell, cm.
W = Weight of dried specimen.
Pb = Percent basic lead styphnate found in barium nitrate filtrate.
Pn = Percent normal lead styphnate found in barium nitrate filtrate.

4.4.1.2.3 Tetracene.-The wet residue obtained in the determination of basic lead styphnate shall be washed three times with alcohol, dried in an oven maintained at 60 ± 5°C, for 30 minutes, cooled in a disiccator and weighed. The dried residue in the crucible shall be transferred to a 125-ml beaker with a stream of water, 25-ml of water shall be added, and the contents of the beaker boiled on a hot plate for 5 minutes. The contents of the beaker shall be poured into the same crucible, and the residue in the crucible washed three times with boiling water and finally with alcohol. The crucible and contents shall be dried in an oven maintained at 60 ± 5°C for 30 minutes, cooled in a desiccator and weighed. The percentage of tetracene in the specimen shall be calculated as follows:

$$\text{Tetracene, percent} = \frac{(C-D)\ 100}{W}$$

where:

C = Weight of crucible and residue after extraction with saturated ammonium acetate solution.
D = Weight of crucible and residue after treating with hot water.
W = Weight of the dried specimen.

4.4.1.2.4 Antimony sulfide.-The weight of the residue shall be calculated from the determination of tetracene to percentage antimony sulfide in the specimen as follows:

$$\text{Antimony sulfide, percent} = \frac{(D-E)\ 100}{W}$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

where:

D = Weight of crucible and specimen after treating with hot water.
E = Weight of crucible.
W = Weight of dried specimen.

4.4.1.2.5 Lead azide.-The lead azide content in the specimen shall be determined by subtracting from 100 the combined percentage of barium nitrate, basic or normal lead styphnate, tetracene and antimony sulfide.

4.4.2 Functioning.

4.4.2.1 Functioning test.-The detonator assemblies shall be mounted in a test fixture. A weight of 0.25 ± 0.01 ounce shall be dropped from a height of three inches on the firing pin. Satisfactory functioning shall be evidenced by complete destruction of detonator.

4.4.2.2 Waterproofness.-The detonators shall be immersed in water to a depth of not less than two inches but not more than three inches for a period of 48 hours, minimum. Both the water and the detonators shall be within the temperature range of 60° to 80° Fahrenheit (F) throughout the test. Within one hour after removal from the water, the detonators shall be tested according to the test procedure for functioning (See 4.4.2.1).

4.4.3 Detonator sticking to packaging.-Visual inspection shall be performed to determine if detonator is sticking to onion skin. Then a "transfer block" shall be used to determine if detonators are sticking to inner set-up box. The "transfer block" shall be placed against the spacer in this inner set-up box and the box in turn shall be inverted. Visual inspection shall be performed to determine if detonator is sticking to inner set-up box. Any detonators remaining in the box because of being cocked shall not be considered defects.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C.-The detonators shall be preserved and packaged in accordance with Dwg. 9258182.

5.2 Packing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

5.2.1 Level C.-The detonators, preserved and packaged as specified in 5.1.1, shall be packed in accordance with Dwg. 9258183.

5.3 Marking.-Marking shall be in accordance with Dwgs. 9258182, 9258183, 925184 and Code of Federal Regulations, Title 49 Parts 171-179.

5.4 Shipping.-When components from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types: Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is catalogued and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the categroy and form (per MIL-D-1000) specified in the Contract Data Requirement Lists (DD Form 1423).

The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design.

6.3.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabrication or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Check test for possible deterioration of detonator.-If the total elapsed time between the original acceptance of any detonator lot and the assembly of that lot into the ammunition component with

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

which it is to be issued exceeds two years, or if the detonators have been subjected to adverse conditions for any length of time sincee-previous tests, the detonator lot shall be subjected to, and shall satisfactorily pass the functioning and waterproofness tests specifed in 4.3.3. The tests shall be conducted at Government expense without cost to the Contractor loading the detonator or the contractor assembling the detonator into the item with which it is issued, and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling or storage permitted by either contractor while the detonator lot is under such contractor's jurisdiction.

6.4.1 Functioning.-The sample plan for the test shall be in accordance with 4.3.3.2.

6.4.2 Waterproofness.-The sampling plan for this test shall be in accordance with 4.3.3.3.

6.5 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, Attn: SARPA-QA-A-S.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1375-A050

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
PICATINNY ARSENAL
DOVER, NEW JERSEY 07801

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314



COMMANDER
Picatinny Arsenal
ATTN: SARPA-QA-A
Dover, New Jersey 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:10Z
Check the source to verify that this is the current version before use.